Advanced Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Composite functions with restrictions

If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = \sqrt{x-5}$, which requires $x - 5 \geq 0$, so the domain is $x \geq 5$.

2 [Foundational] • Target: Composite functions with restrictions

If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

$(f \circ g)(x) = f(\frac{1}{x}) = (\frac{1}{x})^2 - 1 = \frac{1}{x^2} - 1$.

3 [Foundational] • Target: Inverse functions with domain restrictions

Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.

$f^{-1}(x) = \sqrt{x}$. The restriction $x \geq 0$ is necessary because $f(x) = x^2$ is not one-to-one over all real numbers (e.g. $f(2) = f(-2) = 4$), so it has no inverse function unless the domain is restricted to make it one-to-one.

4 [Foundational] • Target: End behavior of polynomial functions

Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.

Since the leading term is $-2x^4$ (even degree, negative coefficient), both ends of the graph point downward: as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$.

5 [Foundational] • Target: End behavior of polynomial functions

Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.

Since the leading term is $3x^5$ (odd degree, positive coefficient): as $x \to \infty$, $g(x) \to \infty$; as $x \to -\infty$, $g(x) \to -\infty$.

6 [Foundational] • Target: Rational functions: asymptotes and holes

Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.

Vertical asymptote: where the denominator is zero: $x = 3$. Horizontal asymptote: since numerator and denominator have the same degree, the asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

7 [Foundational] • Target: Rational functions: asymptotes and holes

Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2-4}{x-2}$.

Factor: $f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2$ for $x \neq 2$. There is a hole at $x = 2$, at the point $(2, 4)$ (found by substituting $x = 2$ into the simplified expression $x + 2$).

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Logarithmic functions and their properties**

Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.

$\log_3(9) + \log_3(27) = \log_3(9 \times 27) = \log_3(243) = \log_3(3^5) = 5$.

9 **[Intermediate]** • **Target: Even, odd, and periodic function classification**

Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.

$f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$. Since $f(-x) = f(x)$, the function is even.

10 **[Intermediate]** • **Target: Piecewise-defined functions and continuity**

A function is defined as $f(x) = 3x + 1$ for $x < 2$ and $f(x) = kx - 3$ for $x \geq 2$. Find the value of k that makes f continuous at $x = 2$.

For continuity, the two pieces must agree at $x = 2$: $3(2) + 1 = k(2) - 3$, so $7 = 2k - 3$, giving $2k = 10$, $k = 5$.

11 **[Intermediate]** • **Target: A multi-part function transformation and composition problem**

This question has two parts. Given $f(x) = 2x - 3$ and $g(x) = x^2$. (a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, in simplified form. (b) Find the value of x for which $(f \circ g)(x) = (g \circ f)(x)$.

(a) $(f \circ g)(x) = f(x^2) = 2x^2 - 3$. $(g \circ f)(x) = g(2x - 3) = (2x - 3)^2 = 4x^2 - 12x + 9$. (b) Set equal: $2x^2 - 3 = 4x^2 - 12x + 9$, so $0 = 2x^2 - 12x + 12$, giving $x^2 - 6x + 6 = 0$. Using the quadratic formula: $x = \frac{6 \pm \sqrt{36 - 24}}{2} = \frac{6 \pm \sqrt{12}}{2} = 3 \pm \sqrt{3}$.

12 **[Intermediate]** • **Target: Finding zeros of higher-degree functions graphically and algebraically**

A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

13 **[Intermediate]** • **Target: Symmetry and transformations of rational functions**

Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.

$f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$. Since $f(-x) = -f(x)$, the function is odd, meaning its graph has point symmetry about the origin — rotating the graph 180° about the origin maps it onto itself.

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14 **[Intermediate]** • **Target: Finding a function's formula from a description of transformations**

The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.

Shifting right by 2 replaces x with $x - 2$: $\sqrt{x-2}$. Vertical stretch by 3: $3\sqrt{x-2}$. Shift down 1: $y = 3\sqrt{x-2} - 1$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: A multi-part rational function analysis problem** • **Cross-Lesson: 4.01 Advanced Trigonometry**

This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.

(a) Numerator: $2x^2 - 8 = 2(x^2 - 4) = 2(x - 2)(x + 2)$. Denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. (b) The factor $(x + 2)$ cancels, indicating a hole at $x = -2$ (specifically at the point found by substituting into the simplified expression). The remaining denominator factor $(x - 3)$ gives a vertical asymptote at $x = 3$ (not canceled). (c) Since the numerator and denominator have the same degree (both degree 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

16 **[Challenging]** • **Target: Finding a quadratic function from a table of values** • **Cross-Lesson: 5.01 Pre-Calculus Concepts**

A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a, b, and c.

From $f(0) = 5$: $c = 5$. From $f(1) = 10$: $a + b + c = 10$, so $a + b = 5$. From $f(2) = 19$: $4a + 2b + c = 19$, so $4a + 2b = 14$, i.e. $2a + b = 7$. Subtracting $a + b = 5$ from $2a + b = 7$: $a = 2$, then $b = 5 - 2 = 3$. Function: $f(x) = 2x^2 + 3x + 5$.

17 **[Challenging]** • **Target: Determining if a function has an inverse using the horizontal line test** • **Cross-Lesson: 6.01 Advanced Coordinate Geometry**

Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.

The horizontal line test states that a function has an inverse (is one-to-one) if and only if no horizontal line intersects its graph more than once. For $f(x) = x^3$, every horizontal line $y = k$ intersects the graph exactly once (since every real number has exactly one real cube root), so $f(x) = x^3$ does have an inverse function, namely $f^{-1}(x) = \sqrt[3]{x}$.

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18 [Challenging] • Target: End behavior of polynomial functions • Cross-Lesson: 7.01 Advanced Plane Geometry

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2+1}{x^2-4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x-2)(x+2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x\to\pm\infty} f(x) = \lim_{x\to\pm\infty} \frac{3+1/x^2}{1-4/x^2} = \frac{3}{1} = 3.$ The horizontal asymptote is $y = 3$. Notice tha.

19 [Challenging] • Target: Rational functions: asymptotes and holes • Cross-Lesson: 8.01 Statistics, Counting & Probability

Determine whether the function $g(x) = \frac{x^3}{x^2+4}$ is even, odd, or neither.

Evaluate $g(-x)$ by substituting $-x$ for every occurrence of x:
$g(-x) = \frac{(-x)^3}{(-x)^2+4} = \frac{-x^3}{x^2+4} = -\left(\frac{x^3}{x^2+4}\right) = -g(x)$ Because $g(-x) = -g(x)$ for all $x \in \mathbb{R}$, $g(x)$ is an odd function (symmetric with respect to 180° rotation about the origin).

20 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 1.01 Advanced Number Theory

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

Evaluate each term according to its interval condition: For $x = -3 < 0$: $f(-3) = (-3)^2 = 9$. For $x = 2 \geq 0$: $f(2) = 2(2) + 1 = 5$. Sum: $f(-3) + f(2) = 9 + 5 = \mathbf{14}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Composite functions with restrictions

Describe the end behavior of $f(x) = 4x^4 - x^3$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. $-\infty, -\infty$

d. ∞, ∞ ✓

The leading term is $4x^4$ (even degree, positive coefficient), so as $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$.

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22 [Foundational] • Target: Inverse functions with domain restrictions

Find the vertical asymptote of $f(x) = \frac{x-15}{x+14}$.

a. -13

b. -15

c. 14

d. -14 ✓

The vertical asymptote occurs where the denominator is zero: $x + 14 = 0$, so $x = -14$.

23 [Foundational] • Target: End behavior of polynomial functions

The quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

a. $-2, -1, 1, 2$ ✓

b. $-1, -2, 1, 2$

c. $1, 2$

d. $-2, 2$

Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u-1)(u-4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

24 [Foundational] • Target: Rational functions: asymptotes and holes

The function f is defined by $f(x) = \frac{ax+8}{cx+d}$ for all real numbers $xeq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. **10** ✓

b. 16

c. -2

d. 2

We determine the constants a, c, and d step-by-step: 1. **Use the x-intercept:** The point $(-4, 0)$ lies on the graph, meaning $f(-4) = 0$. For a rational function, the numerator must equal zero at an x-intercept: $a(-4) + 8 = 0 \Rightarrow -4a = -8 \Rightarrow a = 2$ 2. **Use the horizontal asymptote:** For $f(x) = \frac{ax+8}{cx+d}$, as $x \to \pm\infty$, the horizontal asymptote is given by the ratio of the leading coefficients: $y = \frac{a}{c} = 2$ Since $a = 2$, we have.

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25 [Foundational] • Target: Logarithmic functions and their properties

A function f is defined for all real numbers x such that $x \neq 0$ and $x \neq 1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$ ✓

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

To find $f(3)$, we use the cyclic substitution property of the transformation $T(x) = \frac{1}{1-x}$. 1. Substitute $x = 3$ into the given functional equation: $f(3) + 2f\left(\frac{1}{1-3}\right) = 3(3) \Rightarrow f(3) + 2f\left(-\frac{1}{2}\right) = 9 \quad - - - \text{(Equation 1)}$ 2. Next, evaluate the equation at $x = -\frac{1}{2}$: $\frac{1}{1-(-1/2)} = \frac{1}{3/2} = \frac{2}{3}$.

26 [Foundational] • Target: Even, odd, and periodic function classification

A piecewise function is defined by $f(x) = 4x + 2$ for $x < -3$, and $f(x) = 3x - 4$ for $x \geq -3$. Find $f(-4)$.

a. -15

b. -17

c. **-14** ✓

d. -13

Since $-4 < -3$, use the first piece: $f(-4) = 4(-4) + 2 = -14$.

27 [Foundational] • Target: Piecewise-defined functions and continuity

A polynomial of even degree with positive leading coefficient has end behavior:

a. $y \to -\infty$ as $x \to \pm\infty$

b. $y \to \infty$ as $x \to +\infty$ and $y \to -\infty$ as $x \to -\infty$

c. $y \to +\infty$ **as** $x \to \pm\infty$ ✓

d. Constant

Even degree, positive leading coeff: up on both ends.

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28 [Foundational] • Target: A multi-part function transformation and composition problem

A function has zeros at $x = -1, 0, 2$ and positive leading coefficient. Sketch shows which end behavior?

a. Left: $f \to +\infty$; Right: $f \to -\infty$

b. **Left: $f \to -\infty$; Right: $f \to +\infty$** ✓

c. Both ends to $+\infty$

d. Both ends to $-\infty$

Odd degree (3), positive leading coefficient.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding zeros of higher-degree functions graphically and algebraically

The function f is defined for all real numbers $xeq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7 ✓

By definition of the inverse function, if $f^{-1}(3) = x$, then $f(x) = 3$. Set the function equal to 3 and solve for x: $\frac{4x-1}{x+2} = 3$ Multiply both sides by $x + 2$: $4x - 1 = 3(x + 2)$ $4x - 1 = 3x + 6$ Subtract $3x$ from both sides and add 1: $4x - 3x = 6 + 1$ $x = 7$ Thus, $f^{-1}(3) = 7$. Common Misconceptions: - Choice A (-5) arises from a sign error when clearing the denominator as $3(x - 2)$, yielding $4x - 1 = 3x - 6 \Rightarrow x = -5$. - Choice B (3) results from forgetting to dist.

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30 [Intermediate] • Target: Function operations: sum, difference, product, quotient

The function f is defined by $f(x) = -x^2 + bx + c$, where b and c are constants. In the standard (x, y) coordinate plane, the maximum value of $f(x)$ is 14, and $f(0) = 5$. If $b > 0$, what is the value of $f(5)$?

a. 5

b. 10 ✓

c. 13

d. 14

We are given $f(x) = -x^2 + bx + c$. Step 1: Determine the constant c. Evaluating at $x = 0$: $f(0) = -(0)^2 + b(0) + c = 5 \Rightarrow c = 5$ Step 2: Express the maximum value in terms of b. Since the coefficient of x^2 is -10, we have $b = 6$. Step 4: Evaluate $f(5)$. The complete function is $f(x) = -x^2 + 6x + 5$. $f(5) = -(5)^2 + 6(5) + 5 = -25 + 30 + 5 = 10$ A common error is confusing the parameter b with the vertex coordinate x_v (setting $x_v = 6$), which incorrectly leads to.

31 [Intermediate] • Target: Symmetry and transformations of rational functions

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. **Limit at a is 3,** $\lim_{x \to b^-} f(x) = -\infty$ ✓

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

First, we factor the numerator and the denominator of $f(x) = \frac{x^3 - 8}{x^2 - 4}$. The numerator is a difference of cubes: $x^3 - 8 = (x - 2)(x^2 + 2x + 4)$. The denominator is a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Thus, $f(x) = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)}$. A removable discontinuity occurs where a factor cancels out. Here, $(x - 2)$ cancels, so the removable discontinuity is at $x = a = 2$. To find the limit as x approaches a, we consider the simplified function for $xeq2$.

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32 [Intermediate] • Target: Finding a function's formula from a description of transformations

A function f is defined for all real numbers except $x = 0$ and $x = 1$. For all values of x in its domain, f satisfies the functional equation $f(x) + f\left(\frac{x-1}{x}\right) = 1 + x$. What is the value of $f(2)$?

a. $\frac{3}{4}$ ✓

b. $\frac{5}{4}$

c. $\frac{7}{4}$

d. $\frac{9}{4}$

To find $f(2)$, substitute strategic values of x to generate a system of equations involving the cyclic transformation $g(x) = \frac{x-1}{x}$. 1. Substitute $x = 2$: $\frac{2-1}{2} = \frac{1}{2}$ $f(2) + f\left(\frac{1}{2}\right) = 1 + 2 = 3 \quad - - -$ (Equation 1) 2. Substitute $x = \frac{1}{2}$: $\frac{\frac{1}{2}-1}{\frac{1}{2}} = \frac{-\frac{1}{2}}{\frac{1}{2}} = -1$ $f\left(\frac{1}{2}\right) + f(-1) = 1 + \frac{1}{2} = \frac{3}{2} \quad - - -$ (Equation 2) 3. Substitute $x = -1$.

33 [Intermediate] • Target: A multi-part rational function analysis problem

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. **2** ✓

c. 3

d. 4

For a rational function $f(x) = \frac{P(x)}{Q(x)}$, vertical asymptotes occur at the zeros of the denominator $Q(x)$ that are not also zeros of the numerator $P(x)$. Given a vertical asymptote at $x = 1$, we know that $x = 1$ is a root of the denominator $x^2 - bx + 2 = 0$. Therefore, $1^2 - b(1) + 2 = 0$, which simplifies to $1 - b + 2 = 0$, so $b = 3$. The denominator is $x^2 - 3x + 2 = (x - 1)(x - 2)$. The horizontal asymptote is determined by the ratio of the leading coefficients of the numerator and denominator if t.

34 [Intermediate] • Target: Finding a quadratic function from a table of values

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. **a = 5, b = 3** ✓

b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

For a rational function to have a removable discontinuity at $x = c$, the numerator must have a factor of $(x - c)$ that cancels with a factor in the denominator. Given a removable discontinuity at $x = 2$, the numerator $x^2 - ax + 6$ must have $x = 2$ as a root. Thus, $2^2 - a(2) + 6 = 0$, which simplifies to $4 - 2a + 6 = 0$, so $10 - 2a = 0$, and $a = 5$. The numerator can be factored as $(x - 2)(x - 3)$ since $2 \times 3 = 6$ and $2 + 3 = 5$, so $a = 5$. For the function to have a vertical asymptote at $x = 3$, the.

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35 [Intermediate] • Target: Determining if a function has an inverse using the horizontal line test

The function f is defined by $f(x) = a \cdot b^x + c$, where a, b, and c are constants with $a > 0$, $b > 0$, and $beq1$. In the xy-plane, the horizontal asymptote of the graph of $y = f(x)$ is the line $y = -4$. If the graph passes through the points $(0, -1)$ and $(2, 8)$, what is the value of $f(3)$?

a. **20** ✓

b. 24

c. 32

d. 44

Step 1: Determine the vertical shift parameter c. For an exponential function in the form $f(x) = a \cdot b^x + c$, the term $a \cdot b^x$ approaches 0 as $x \to -\infty$ (for $b > 1$) or as $x \to \infty$ (for 00), we find $b = 2$. Step 4: Evaluate $f(3)$. The complete function is $f(x) = 3 \cdot 2^x - 4$. Evaluating at $x = 3$:
$f(3) = 3 \cdot 2^3 - 4 = 3(8) - 4 = 24 - 4 = 20$ Misconception Traps: - Choice B (24) results from computing $a \cdot b^3 = 3 \cdot 8 = 24$ but forgetting to subt.

36 [Intermediate] • Target: Composite functions with restrictions

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?

a. 24 ✓

b. 18

c. 17

d. 14

We solve the problem in two key phases: identifying the polynomial $f(x)$ from its graphical features, and applying the Polynomial Remainder Theorem to determine $R(x)$. **Step 1: Determine $f(x)$** - Because the graph is tangent to the x-axis at $x = -2$, $(x + 2)$ is a factor with an even multiplicity of at least 2. - Because the graph crosses the x-axis at $x = 1$ and $x = 3$, $(x - 1)$ and $(x - 3)$ are factors with odd multiplicities of at least 1. - Since $f(x)$ is of degree 4, the sum of multipliciti.

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37 [Intermediate] • Target: Inverse functions with domain restrictions

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?

a. $8 - 2\sqrt{6}$ ✓

b. 4

c. $\frac{10}{3}$

d. $2\sqrt{6}$

1. **Find the total area of ΔABC:** The base along the x-axis connects $A(0, 0)$ and $B(8, 0)$, so the base length is $b = 8$. The vertex $C(2, 6)$ gives a height of $h = 6$. Area$(\Delta ABC) = \frac{1}{2} \times 8 \times 6 = 24$ Each of the two regions must therefore have an area of $\frac{24}{2} = 12$. 2. **Determine the location of $x = k$:** The vertical line through C is $x = 2$. The portion of ΔABC to the left of $x = 2$ is a triangle with base on the x-axis from $x = 0$ to.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: End behavior of polynomial functions • Cross-Lesson: 7.01 Advanced Plane Geometry

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?

a. $-\frac{5}{2}$ ✓

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

To determine $f(-1)$, we first find the constants a, b, and c: 1. **Horizontal Asymptote:** For an exponential function of the form $f(x) = a \cdot b^x + c$, as the exponential term approaches zero, the function approaches its horizontal asymptote $y = c$. Given that the asymptote is $y = -4$, we have: $c = -4$ 2. **Use the Point $(0, -1)$:** Substitute $x = 0$ and $f(0) = -1$: $f(0) = a \cdot b^0 - 4 = -1$ $a(1) - 4 = -1 \Rightarrow a = 3$ 3. **Use the Point $(2, 8)$:** Substitute $x = 2$ and $f(2) = 8$.

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39 [Challenging] • Target: Rational functions: asymptotes and holes • Cross-Lesson: 8.01 Statistics, Counting & Probability

The graph of the quadratic function $f(x) = ax^2 + bx + c$ is shown in the standard (x, y) coordinate plane below, with vertex $(2, -1)$ and y-intercept $(0, 3)$. The function g is defined by $g(x) = f(x + k) - 8$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, which of the following could be the value of k?

a. -5

b. -3

c. 1

d. 5 ✓

Step 1: Find the equation of $f(x)$. Using the vertex form of a parabola with vertex $(h, k_v) = (2, -1)$: $f(x) = a(x - 2)^2 - 1$ Since the graph passes through the y-intercept $(0, 3)$, substitute $x = 0$ and $y = 3$: $3 = a(0 - 2)^2 - 1 \Rightarrow 3 = 4a - 1 \Rightarrow 4a = 4 \Rightarrow a = 1$ Thus, $f(x) = (x - 2)^2 - 1$. Step 2: Apply the condition that g passes through $(0, 0)$. Since $g(0) = 0$ and $g(x) = f(x + k) - 8$: $g(0) = f(0 + k) - 8 = f(k) - 8 = 0 \Rightarrow f(k) = 8$ Step 3: Solve for k..

40 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 1.01 Advanced Number Theory

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?

a. -4

b. -2 ✓

c. -1

d. 1

We solve the problem in three steps: 1. **Determine the quadratic function $f(x)$:** The vertex is given as $(h, k) = (2, 9)$, so the function has the form: $f(x) = a(x - 2)^2 + 9$ Since the graph passes through the point $(0, 5)$: $5 = a(0 - 2)^2 + 9 \Rightarrow 4a + 9 = 5 \Rightarrow 4a = -4 \Rightarrow a = -1$ Expanding $f(x)$: $f(x) = -(x - 2)^2 + 9 = -(x^2 - 4x + 4) + 9 = -x^2 + 4x + 5$ 2. **Determine the linear function $g(x)$:** The line $g(x) = mx + 1$ passes through the vertex $(2, 9)$.

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41 [Challenging] • Target: Even, odd, and periodic function classification • Cross-Lesson: 2.01 Advanced Algebra & Systems

The function f is defined by $f(x) = \frac{2x^2 + kx - 6}{x - 2}$ for all real numbers $xeq2$, where k is a constant. If the graph of $y = f(x)$ in the standard (x, y) coordinate plane has a removable discontinuity (a hole) at the point $(2, L)$, what is the value of L?

a. -1

b. 3

c. 7 ✓

d. 11

A rational function has a removable discontinuity at $x = 2$ if and only if both the numerator and the denominator share the factor $(x - 2)$. Therefore, the numerator must equal 0 when $x = 2$: $2(2)^2 + k(2) - 6 = 0\ 8 + 2k - 6 = 0\ 2k + 2 = 0 \Rightarrow k = -1$ Now, substitute $k = -1$ back into the numerator and factor it: $2x^2 - x - 6 = (x - 2)(2x + 3)$ For all $xeq2$, simplify $f(x)$: $f(x) = \frac{(x - 2)(2x + 3)}{x - 2} = 2x + 3$ The y-coordinate L of the removable discontinuity is the limit of.

42 [Challenging] • Target: Piecewise-defined functions and continuity • Cross-Lesson: 4.01 Advanced Trigonometry

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?

a. -10

b. -6

c. 6

d. **10** ✓

Step 1: Write the equation of $f(x)$ in vertex form. Since the vertex is $(h, k_v) = (1, 8)$, the quadratic function has the form: $f(x) = a(x - 1)^2 + 8$ Step 2: Use the known point $(3, 0)$ to solve for a. $f(3) = a(3 - 1)^2 + 8 = 0 \Rightarrow 4a + 8 = 0 \Rightarrow a = -2$ Thus, $f(x) = -2(x - 1)^2 + 8$. Step 3: Apply the condition that $g(x)$ passes through $(0, 0)$. $g(0) = f(0 - 2) + k = 0 \Rightarrow f(-2) + k = 0$ Step 4: Evaluate $f(-2)$. $f(-2) = -2(-2 - 1)^2 + 8 = -2(-3)^2 + 8 = -2(9) + 8 = -10$ Step.

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43 [Challenging] • Target: A multi-part function transformation and composition problem • Cross-Lesson: 5.01 Pre-Calculus Concepts

The functions f and g are defined for all real numbers by $f(x) = 2x^2 - 3$ and $g(x) = 3x + 1$ What is the value of $f(g(-2))$?

a. -53

b. 16

c. 47 ✓

d. 95

To find the composite value $f(g(-2))$, evaluate the inner function first, then substitute the result into the outer function: 1. Evaluate $g(-2)$: $g(-2) = 3(-2) + 1 = -6 + 1 = -5$ 2. Substitute -5 into $f(x)$: $f(-5) = 2(-5)^2 - 3 = 2(25) - 3 = 50 - 3 = 47$

44 [Challenging] • Target: Finding zeros of higher-degree functions graphically and algebraically • Cross-Lesson: 6.01 Advanced Coordinate Geometry

A line L is tangent to the graph of the polynomial function $f(x) = x^4 - 6x^2 + 5x + 7$ at two distinct points in the xy-plane. What is the y-intercept of line L?

a. **-2** ✓

b. 2

c. 7

d. 16

Let the equation of the tangent line L be $y = mx + b$. Because L is tangent to the quartic polynomial $f(x)$ at two distinct points, say $x = r$ and $x = s$ with res, each point of tangency corresponds to a root of multiplicity at least 2 for the difference equation $f(x) - (mx + b) = 0$. Since $f(x) - (mx + b)$ is a monic polynomial of degree 4, it factors completely over the real numbers as: $f(x) - (mx + b) = (x - r)^2(x - s)^2$ Expanding this product.

45 [Challenging] • Target: Function operations: sum, difference, product, quotient • Cross-Lesson: 7.01 Advanced Plane Geometry

The polynomial function $P(x) = x^3 + ax^2 + bx + 12$, where a and b are real constants, has $x - 1$ as a factor. When $P(x)$ is divided by $x - 2$, the remainder is -10. What is the greatest root of the equation $P(x) = 0$?

a. **4** ✓

b. 3

c. 1

d. 6

To find the roots of $P(x) = 0$, we first determine the unknown parameters a and b using the Factor Theorem and the Remainder Theorem. 1. **Apply the Factor Theorem:** Since $x - 1$ is a factor of $P(x)$, we have $P(1) = 0$: $P(1) = (1)^3 + a(1)^2 + b(1) + 12 = 0 \Rightarrow 1 + a + b + 12 = 0 \Rightarrow a + b = -13$ 2. **Apply the Remainder Theorem:** When $P(x)$ is divided by $x - 2$, the remainder is -10, so $P(2) = -10$.

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